

CONEXÃO LOCAL COM AWS, AZURE E GCP

- GUIA COMPLETO

CONFIGURAÇÃO INICIAL E CLI's

Instalação das CLI's

```
bash
```

```
# AWS CLI
pip install awscli
# ou
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
unzip awscliv2.zip
sudo ./aws/install
```

```
# Azure CLI
curl -sL https://aka.ms/InstallAzureCLIDeb | sudo bash
# ou no macOS
brew update && brew install azure-cli
```

```
# Google Cloud CLI
curl https://sdk.cloud.google.com | bash
```

```
exec -l $SHELL
```

AWS - CONFIGURAÇÃO E CONEXÃO

Configuração Inicial AWS

```
bash
```

```
# Configurar credenciais
aws configure
# Ou setar variáveis de ambiente
export AWS_ACCESS_KEY_ID=AKIAIOSFODNN7EXAMPLE
```

```
export AWS_SECRET_ACCESS_KEY=wJa1rXUtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY
export AWS_DEFAULT_REGION=us-east-1
```

```
# Verificar configuração
aws sts get-caller-identity
```

```
aws configure list
```

Comandos Essenciais AWS

```
bash
```

```
# EC2
aws ec2 describe-instances
aws ec2 start-instances --instance-ids i-1234567890abcdef0
aws ec2 terminate-instances --instance-ids i-1234567890abcdef0
```

```
# S3
aws s3 ls
aws s3 cp arquivo.txt s3://meu-bucket/
aws s3 sync ./local-folder s3://meu-bucket/
```

```
# IAM
aws iam list-users
aws iam create-user --user-name novo-usuario
```

```
# CloudFormation
```

```
aws cloudformation create-stack --stack-name minha-stack
--template-body file://template.yaml
```

Conexão SSH com Instâncias EC2

```
bash
```

```
# Conectar via SSH
ssh -i "minha-key.pem" ec2-user@ec2-xx-xx-xx-xx.compute-1.amazonaws.com
```

```
# Configurar SSH no ~/.ssh/config
Host aws-ec2
    HostName ec2-xx-xx-xx-xx.compute-1.amazonaws.com
    User ec2-user
```

```
IdentityFile ~/.ssh/minha-key.pem
```

```
# Conectar com alias
```

```
ssh aws-ec2
```

Session Manager (Conexão sem SSH)

```
bash
```

```
# Conectar via Session Manager
```

```
aws ssm start-session --target i-1234567890abcdef0
```

```
# Instalar Session Manager plugin
```

```
# Download:
```

```
https://docs.aws.amazon.com/systems-manager/latest/userguide/session-manager-working-with-install-plugin.html
```



AZURE - CONFIGURAÇÃO E CONEXÃO

Configuração Inicial Azure

```
bash
```

```
# Login
```

```
az login
```

```
az login --use-device-code # Para dispositivos sem browser
```

```
# Logout
```

```
az logout
```

```
# Verificar Login
```

```
az account show
```

```
az account list
```

```
# Setar subscription
```

```
az account set --subscription "Minha Subscription"
```

```
# Configurar default values
```

```
az configure --defaults group=MeuResourceGroup location=brazilsouth
```

Comandos Essenciais Azure

```
bash
```

Resource Groups

```
az group create --name MeuRG --location eastus  
az group list
```

Virtual Machines

```
az vm create --resource-group MeuRG --name MinhaVM --image UbuntuLTS  
--admin-username azureuser  
az vm list  
az vm start --resource-group MeuRG --name MinhaVM  
az vm stop --resource-group MeuRG --name MinhaVM
```

Storage

```
az storage account create --resource-group MeuRG --name mystorageaccount  
--location eastus --sku Standard_LRS  
az storage blob list --account-name mystorageaccount --container-name meucontainer
```

App Service

```
az webapp create --resource-group MeuRG --plan MeuAppServicePlan  
--name MeuWebApp
```

Conexão com VMs Azure

```
bash
```

Conectar via SSH

```
ssh azureuser@52.191.120.100
```

Usando Azure Bastion (via browser)

Acessar portal.azure.com -> VM -> Connect -> Bastion

Conectar com Azure CLI

```
az ssh vm --name MinhaVM --resource-group MeuRG
```

Reset de senha SSH

```
az vm user update --username azureuser --password Nov@Senh@123
--resource-group MeuRG --name MinhaVM
```



GOOGLE CLOUD - CONFIGURAÇÃO E CONEXÃO

Configuração Inicial GCP

```
bash
```

Login

```
gcloud auth login
```

```
gcloud auth login --no-launch-browser # Para obter URL manual
```

Listar projetos

```
gcloud projects list
```

Setar projeto

```
gcloud config set project MEU-PROJETO-ID
```

Verificar configuração

```
gcloud config list
```

```
gcloud auth list
```

Configurar compute zone

```
gcloud config set compute/zone us-central1-a
```

Comandos Essenciais GCP

```
bash
```

Compute Engine

```
gcloud compute instances list
```

```
gcloud compute instances create minha-instancia --machine-type n1-standard-1
--image-family ubuntu-2004-lts --image-project ubuntu-os-cloud
```

```
gcloud compute instances start minha-instancia
```

```
gcloud compute instances stop minha-instancia
```

Storage

```
gsutil ls
```

```
gsutil cp arquivo.txt gs://meu-bucket/  
gsutil rsync -r ./local-folder gs://meu-bucket/
```

Kubernetes Engine

```
gcloud container clusters list
```

```
gcloud container clusters get-credentials meu-cluster --zone  
us-central1-a
```

Conexão com Instâncias GCP

```
bash
```

Conectar via SSH

```
gcloud compute ssh minha-instancia --zone us-central1-a
```

Conectar com porta específica

```
gcloud compute ssh minha-instancia --zone us-central1-a --ssh-flag="-L  
8080:localhost:8080"
```

Usando OS Login (recomendado)

```
gcloud compute instances add-metadata minha-instancia --metadata  
enable-oslogin=TRUE
```

Conectar via browser (Google Cloud Console)

```
# Acessar console.cloud.google.com -> Compute Engine -> VM instances  
-> SSH
```



GERENCIAMENTO MULTI-CLOUD

Configuração de Múltiplas Clouds

```
bash
```

Perfil AWS

```
aws configure --profile dev  
aws configure --profile prod  
aws s3 ls --profile dev
```

```
# Configurações Azure
```

```
az account list --output table
```

```
az config get defaults
```

```
# Configurações GCP
```

```
gcloud config configurations list
```

```
gcloud config configurations create dev
```

```
gcloud config configurations activate prod
```

Script de Conexão Multi-Cloud

```
bash
```

```
#!/bin/bash
```

```
# cloud-connect.sh
```

```
case $1 in
```

```
  "aws")
```

```
    echo "Conectando à AWS..."
```

```
    ssh -i ~/.ssh/aws-key.pem ec2-user@$2
```

```
    ;;
```

```
  "azure")
```

```
    echo "Conectando ao Azure..."
```

```
    ssh azureuser@$2
```

```
    ;;
```

```
  "gcp")
```

```
    echo "Conectando ao GCP..."
```

```
    gcloud compute ssh $2
```

```
    ;;
```

```
  *)
```

```
    echo "Uso: $0 {aws|azure|gcp} [host]"
```

```
    ;;
```

```
esac
```



FERRAMENTAS DE TERCEIROS

Terraform (Infrastructure as Code)

```
bash
```

```
# Arquivo main.tf para multi-cloud
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 4.0"
    }
    azurearm = {
      source = "hashicorp/azurearm"
      version = "~> 3.0"
    }
    google = {
      source = "hashicorp/google"
      version = "~> 4.0"
    }
  }
}
```

```
# Comandos Terraform
terraform init
terraform plan
terraform apply
```

```
terraform destroy
```

Exemplo Terraform Multi-Cloud

```
hcl
```

```
# provider.tf
provider "aws" {
  region = "us-east-1"
}

provider "azurearm" {
  features {}
}

provider "google" {
  project = "meu-projeto"
  region = "us-central1"
}

# Criar instância em cada cloud
resource "aws_instance" "web" {
  ami          = "ami-12345678"
  instance_type = "t2.micro"
```

```

}

resource "azurerm_linux_virtual_machine" "web" {
  name                = "vm-azure"
  resource_group_name = azurerm_resource_group.example.name
  location            = azurerm_resource_group.example.location
  size                = "Standard_B1s"
  admin_username      = "adminuser"

  network_interface_ids = [azurerm_network_interface.example.id]

  admin_ssh_key {
    username   = "adminuser"
    public_key = file("~/ssh/id_rsa.pub")
  }

  os_disk {
    caching              = "ReadWrite"
    storage_account_type = "Standard_LRS"
  }

  source_image_reference {
    publisher = "Canonical"
    offer     = "UbuntuServer"
    sku       = "16.04-LTS"
    version   = "latest"
  }
}

resource "google_compute_instance" "web" {
  name         = "vm-gcp"
  machine_type = "e2-micro"
  zone         = "us-central1-a"

  boot_disk {
    initialize_params {
      image = "ubuntu-os-cloud/ubuntu-2004-lts"
    }
  }

  network_interface {
    network = "default"
    access_config {}
  }
}

```



SEGURANÇA E BOAS PRÁTICAS

Gerenciamento de Credenciais

```
bash
```

```
# Usando AWS Vault (para AWS)
```

```
brew install aws-vault
```

```
aws-vault add myprofile
```

```
aws-vault exec myprofile -- aws s3 ls
```

```
# Azure Managed Identity
```

```
az login --identity
```

```
# GCP Service Accounts
```

```
gcloud auth activate-service-account --key-file=service-account-key.json
```

```
# Variáveis de ambiente seguras
```

```
export AWS_ACCESS_KEY_ID=""
```

```
export AWS_SECRET_ACCESS_KEY=""
```

```
export AZURE_CLIENT_ID=""
```

```
export AZURE_CLIENT_SECRET=""
```

```
export
```

```
GOOGLE_APPLICATION_CREDENTIALS="path/to/service-account-key.json"
```

MFA (Multi-Factor Authentication)

```
bash
```

```
# AWS MFA
```

```
aws sts get-session-token --serial-number arn:aws:iam::123456789012:mfa/user
```

```
--token-code 123456
```

```
# Azure MFA (via browser)
```

```
az login # Irá solicitar MFA automaticamente
```

```
# GCP 2-Step Verification
```

```
gcloud auth login # Suporte nativo a 2FA
```



MONITORAMENTO MULTI-CLOUD

Comandos de Monitoramento

```
bash
```

```
# AWS CloudWatch
```

```
aws cloudwatch list-metrics --namespace AWS/EC2
```

```
aws cloudwatch get-metric-statistics --namespace AWS/EC2 --metric-name  
CPUUtilization
```

```
# Azure Monitor
```

```
az monitor metrics list-definitions --resource
```

```
/subscriptions/{sub}/resourceGroups/{rg}/providers/Microsoft.Compute/virtualMachines/{vm}
```

```
az monitor activity-log list
```

```
# Google Cloud Monitoring
```

```
gcloud monitoring metrics list
```

```
gcloud logging read "resource.type=gce_instance"
```



SCRIPTS DE AUTOMAÇÃO

Script de Health Check Multi-Cloud

```
bash
```

```
#!/bin/bash
```

```
# cloud-health-check.sh
```

```
echo "=== HEALTH CHECK MULTI-CLOUD ==="
```

```
# AWS Health
```

```
echo -e "\n💠 AWS Status:"
```

```
aws ec2 describe-instances --query
```

```
'Reservations[].Instances[].{ID:InstanceId,State:State.Name}' --output table
```

```
# Azure Health
```

```
echo -e "\n🟦 Azure Status:"
```

```
az vm list -d --query "[].{Name:name,State:powerState}" -o table
```

```
# GCP Health
```

```
echo -e "\n🟢 GCP Status:"
```

```
gcloud compute instances list --format="table(name,status)"
```

Script de Backup Multi-Cloud

```
bash
```

```
#!/bin/bash
```

```
# multi-cloud-backup.sh
```

```
BACKUP_DATE=$(date +%Y%m%d_%H%M%S)
```

```
echo "Iniciando backup multi-cloud: $BACKUP_DATE"
```

```
# AWS S3 Backup
```

```
echo "Fazendo backup AWS S3..."
```

```
aws s3 sync /dados/important-files s3://my-backup-bucket/$BACKUP_DATE/
```

```
# Azure Blob Storage Backup
```

```
echo "Fazendo backup Azure Blob Storage..."
```

```
az storage blob upload-batch --destination mycontainer --source  
/dados/important-files --pattern "*"
```

```
# Google Cloud Storage Backup
```

```
echo "Fazendo backup Google Cloud Storage..."
```

```
gsutil -m rsync -r /dados/important-files gs://my-backup-bucket/$BACKUP_DATE/
```

```
echo "Backup concluído!"
```



TROUBLESHOOTING

Problemas Comuns e Soluções

```
bash
```

```
# AWS - Credenciais expiradas
```

```
aws sts get-caller-identity # Verifica se credenciais são válidas
```

```
# Azure - Subscription não encontrada
```

```
az account list --output table # Lista todas as subscriptions
```

GCP - Projeto não setado

```
gcloud config set project MEU-PROJETO-ID
```

Problemas de rede

```
curl -I https://sts.amazonaws.com/ # Testa conectividade AWS
```

```
curl -I https://management.azure.com/ # Testa conectividade Azure
```

```
curl -I https://www.googleapis.com/ # Testa conectividade GCP
```

Timeout em operações

```
aws configure set default.socket_timeout 300
```

```
az config set core.http_timeout=300
```

```
gcloud config set core/connect_timeout 300
```

Verificação de Permissões

```
bash
```

AWS IAM

```
aws iam list-attached-user-policies --user-name meu-usuario
```

Azure RBAC

```
az role assignment list --assignee meu-usuario@dominio.com
```

GCP IAM

```
gcloud projects get-iam-policy MEU-PROJETO-ID
```

Este guia fornece todas as ferramentas necessárias para conectar e gerenciar recursos nas três principais clouds! ☁️